An Agentic AI Framework for Intelligent Patient Monitoring and Clinical Decision Support with Patient-Timeline Retrieval and Verified Recommendations

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***Abstract*—Clinicians reason over a patient's trajectory; the large language model (LLM) agents now entering clinical workflows reason over a prompt. Although such agents can invoke validated tools, sustain diagnostic dialogue, and in one documented case run prospectively in silent mode in a live clinical setting, we identify three properties needed for trust in a specific patient that, to the best of our knowledge, no published system measures in combination: retrieval grounded in the patient's own longitudinal record rather than external knowledge alone, evaluation on real intensive-care admissions rather than examinations or synthetic records, and verification whose audit trail is itself checked rather than assumed. We present an agentic AI framework for intelligent patient monitoring and clinical decision support that makes each of these a designed, measurable component. Specialized agents for monitoring, diagnosis, risk prediction, treatment recommendation, explanation, and verification operate under a coordinator, over a memory layer in which the patient timeline is a first-class retrieval corpus; recommendations must pass an evidence-entailment verification gate before reaching a structured clinician review step, accompanied by a calibrated confidence estimate. We specify an evaluation design on MIMIC-IV that scores longitudinal patient tracking, and report a pilot feasibility study on the openly licensed 100-patient MIMIC-IV demo: timelines for 140 ICU stays build in under three seconds; timestamp-aware retrieval answers in about 12 ms without returning future information; a rule-based verification gate blocks 86% of false alerts while retaining 43% of true alerts at its strictest informative threshold; and all 183 logged audit-trail references re-resolve exactly against source records. The pilot establishes implementability, not clinical utility; the framework's contribution is making trustworthiness properties measurable that current systems only assert.**

***Index Terms*—agentic AI, clinical decision support, patient monitoring, large language models, retrieval-augmented generation, verification, audit trail, human-in-the-loop, agent interoperability protocols (MCP, A2A), MIMIC-IV.**

I. Introduction

An intensive-care clinician deciding whether to escalate treatment reasons over a patient's trajectory: what the lactate did overnight, which medications were started and stopped, how this admission compares with the last one. The language-model systems now entering clinical workflows do not reason this way. They can select and invoke validated clinical tools [1], [2], sustain multi-turn diagnostic dialogue [3], [4], and in one documented case run prospectively, in silent mode, inside a live emergency-department deployment [5]; their underlying machinery, from reasoning-and-acting loops [6] and retrieval-augmented generation [7] to multi-agent coordination [8] and structured memory [9], is individually mature. Each of these systems, however, rebuilds its picture of the patient from the current prompt and retains nothing once the interaction ends.

This research is motivated by three shortfalls that follow from that mismatch between clinical reasoning and system design. The first concerns evaluation substrate. The strongest medical agents are still graded on curated examinations or synthetic records [10], [11], [12], whereas a real intensive-care admission is noisy, longitudinal, and internally revised as results are corrected and orders change [13]; competence on the former substrate does not establish competence on the latter. The second concerns grounding direction. Clinical retrieval-augmented systems, including those with strong results, treat external knowledge as the corpus and the patient as the query [14], [15]. Emerging work grounds models in the longitudinal record for prediction and summarization, but there the record is input to be compressed, not timestamp-addressable evidence coupled to each recommendation the system makes. The third concerns verifiability. The field can now characterize hallucination and uncertainty in clinical language models [16], [17], and regulators have begun demanding inspectable safeguards for agentic clinical software [18], [19], but published systems rarely check an individual recommendation against the patient evidence that was retrieved for it, and within the literature reviewed for this study we found none that measures whether its own audit trail faithfully records what the system used.

This paper presents an agentic AI framework for intelligent patient monitoring and clinical decision support in which each of those three properties is a designed, measurable component rather than an aspiration. Specialized LLM agents for monitoring, diagnosis, risk prediction, treatment recommendation, explanation, and verification operate under a coordinator. Beneath them, a memory layer makes the patient's longitudinal EHR timeline a first-class retrieval corpus. Above them, a verification gate blocks recommendations that the retrieved evidence does not support, writing every decision to an audit trail whose faithfulness is itself an evaluation target. Clinician oversight is structured rather than advisory; recommendations arrive only after verification, carrying linked evidence and

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Research prototype disclosure: the system described is an academic proof-of-concept evaluated on de-identified and synthetic data only; it is not a medical device and is not intended for clinical use.

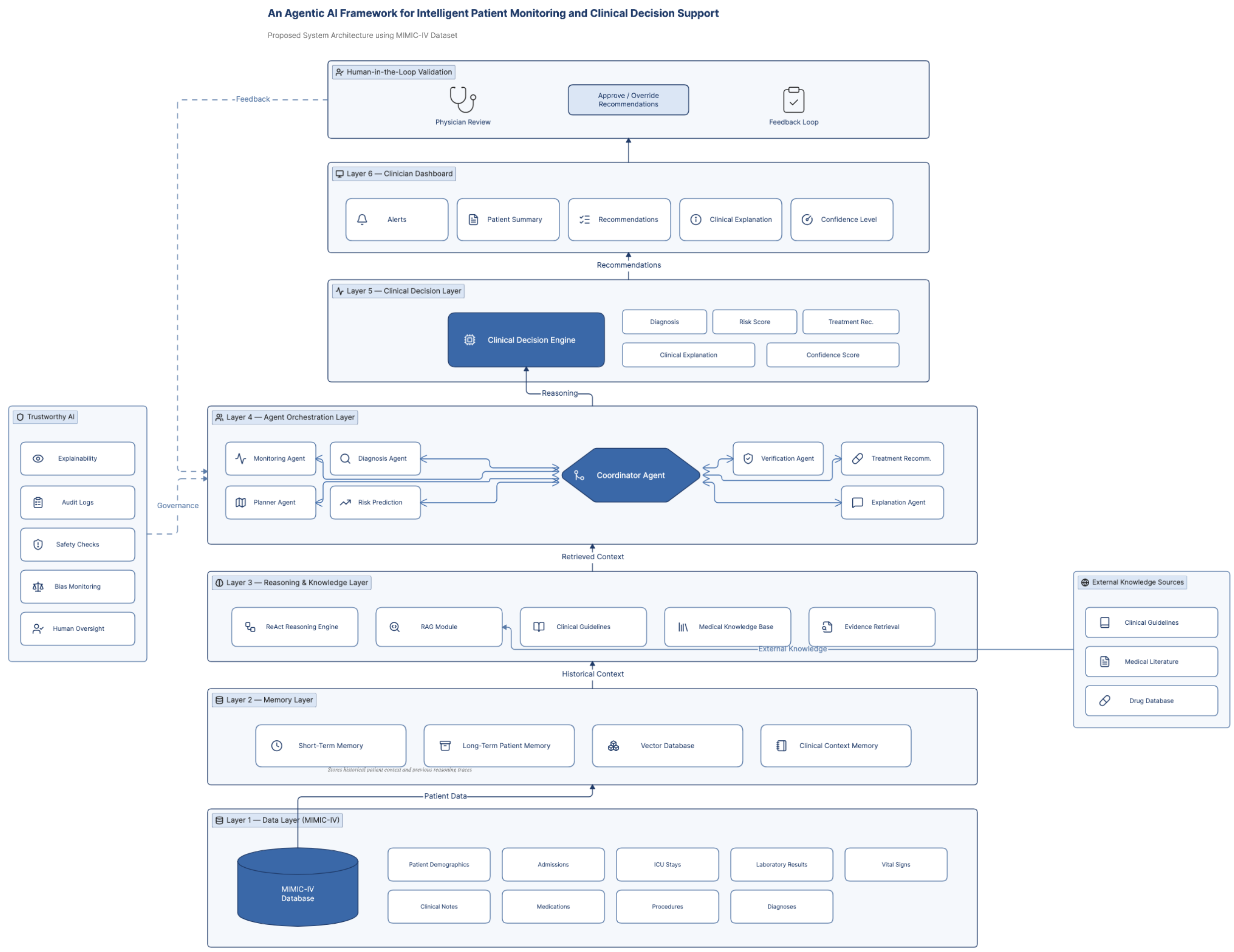

Fig. 1. The proposed agentic AI framework over MIMIC-IV. Six layers (data, memory, reasoning and knowledge, agent orchestration, clinical decision, clinician dashboard) are closed by human-in-the-loop validation; a cross-cutting trustworthy-AI panel (explainability, audit logs, safety checks, bias monitoring, human oversight) governs every layer, and external knowledge sources feed the retrieval module.

a calibrated confidence estimate, because the meta-analytic record suggests unstructured human–LLM collaboration yields gains too uncertain to rely on [20].

The contributions are: (1) a layered framework that grounds retrieval in the patient timeline rather than only in external knowledge; (2) a verification gate and evidence-linked audit trail specified as measurable components, with metrics for recommendation grounding and trail faithfulness; (3) an evaluation design on MIMIC-IV that scores longitudinal patient tracking rather than one-shot question answering, positioned against the exam-based and synthetic-EHR benchmarks that dominate current practice [12], [21], [22]; (4) an analysis of how the framework's safeguards map onto emerging regulatory requirements for unconfined non-deterministic clinical software [18]; and (5) a pilot feasibility study on the openly licensed MIMIC-IV demo that grounds the framework's distinguishing mechanisms in de-identified real-world ICU data: timelines for 140 stays build in under three seconds, timestamp-aware retrieval answers in roughly 12 ms without returning future information, the verification gate's evidence requirement suppresses false alerts faster than true ones (86% versus 57% blocked at four required signals), and every logged audit-trail reference re-resolves against the source record. These are feasibility results on a 100-patient cohort, and the paper reports them with that caveat throughout.

The remainder of the paper reviews related work (Section II), details the framework and its evaluation design (Section III), discusses implications and limitations (Section IV), and reports the pilot study (Section V).

II. Related Work

We organize prior work around four questions that any clinical decision-support agent must answer: how agents are built, what medical agents actually do, what their retrieval is grounded in, and how anyone would know whether to trust them. Reading the literature through these questions, rather than system by system, makes the unoccupied territory visible.

A. How Agents Are Built

The technical substrate of agentic systems accumulated in three recognizable waves. The first gave single agents their core competencies: interleaving reasoning with action [6], acquiring tool use without hand-labeled supervision [23], and, as surveys of the period document, coupling these with memory, planning, and self-reflection into a general architecture [24], [25]. A second wave asked how several such agents should work together, and its answers differ mainly in where coordination logic lives: in conversation itself [8], in assigned social roles [26], or in codified standard operating procedures [27]. The third wave is consolidation. Collaboration patterns now have a formal taxonomy [28], cognitive modules a brain-inspired reference blueprint [29], memory a construction–management–retrieval decomposition [9], and inter-agent communication a set of emerging wire protocols [30]. The distinction between a tool-using agent and an orchestrated agentic ecosystem has itself been formalized, with healthcare cited among the application domains where the orchestrated form matters most [31].

For present purposes the significance of this maturation is double-edged. It means a clinical framework can be assembled from settled components, so integration alone earns no novelty. It also means the open problems are no longer architectural: nothing in this infrastructure literature says what a *clinical* agent should be grounded in, or how its outputs should be verified before they reach a patient. Those questions belong to the application layer, and the infrastructure work is silent on them by design.

B. What Medical Agents Do

Work applying language models to medicine splits into two generations distinguished by where capability resides. In the first, capability is parametric: models encode clinical knowledge [32], answer examination questions at expert level [33], extend across modalities [34], and are reproduced in open weights [35]. The second generation externalizes capability into structure around the model, and the choice of structure is what differentiates otherwise similar systems. Some externalize *deliberation*, convening role-played specialist panels or simulated clinical environments in which diagnostic behavior can develop [10], [11], [36]. Some externalize *record access*, generating executable queries over structured EHR tables [37]. Some externalize *computation*, delegating therapeutic reasoning, risk scoring, or image analysis to hundreds of validated tools rather than trusting generation [1], [2], [38]. Others externalize the *interaction policy* itself, learning when to ask rather than answer [39], or extend dialogue systems toward multi-visit management and multimodal evidence gathering [3], [4].

Viewed together, these design choices share an assumption that none of the papers states: the patient exists only for the duration of a prompt. Whatever is externalized (deliberation, queries, tools, or dialogue policy), patient state is reassembled from the scenario at each step, and none of these systems carries an evidence-linked model of one real patient forward through an admission. The broadest survey of the area finds the research effort concentrated on clinical decision-making while naming implementation barriers among the field's central open challenges [40]. The gap this paper targets sits at the intersection of those two observations.

C. What Retrieval Is Grounded In

Retrieval-augmented generation earned its place as the default hallucination mitigation [7], [41], and its refinements track a single trajectory: giving the model more control over its own evidence, first through self-critique of retrieved passages [42], then by folding planning, iteration, and reflection over retrieval into the agent loop itself [43]. Medical instantiations follow the same trajectory while narrowing the corpus to curated clinical knowledge, whether a diagnostic knowledge graph [14] or perioperative guidelines, where grounding measurably outperformed clinicians in a bounded assessment task [15]. Across this RAG line, the direction of grounding has not varied: external knowledge is the corpus and the patient is the query.

A newer line of work has begun to treat the longitudinal record itself as the substrate, and it deserves separate attention. CliCARE compiles a patient's cancer EHR into a temporal knowledge graph and aligns it with guideline knowledge to generate recommendations, evaluated on MIMIC-IV among other data [44]. Traj-CoA and its successor TrajOnco process multi-year records with chains of summarizing agents whose distilled timelines feed cancer risk prediction [45], [46]. TIMER benchmarks and instruction-tunes temporal reasoning grounded to segments of real longitudinal records [47], and RGAR iteratively refines retrieval between the EHR context attached to a question and external knowledge [48]. These systems establish that longitudinal grounding is both feasible and valuable. They also sharpen, rather than close, the opening this work occupies: in each, the patient record serves prediction, summarization, or question answering as input to be compressed or aligned, not as timestamp-addressable evidence coupled to individual recommendations; none verifies a recommendation against the evidence retrieved for it; none measures the faithfulness of an audit trail; and their evaluations score endpoint accuracy rather than sustained tracking of one admission. What remains unoccupied, within the literature reviewed for this study, is the integration this paper proposes and instruments: timestamp-aware retrieval from the patient timeline, coupled to recommendation-level verification and a measured audit trail, evaluated longitudinally on de-identified real-world ICU records.

D. How Trust Would Be Established

Whether these systems can be trusted is being addressed by three literatures that do not yet meet. Evaluation research is moving benchmarks toward realism, placing agents inside a FHIR-compliant virtual EHR with physician-authored tasks [12], grading open-ended conversations against physician-written rubrics [21], and cataloguing which agent capabilities current benchmarks measure, with safety flagged as the persistent omission [49]. On de-identified real-world records, the evidence is thinner and more sobering: revisited MIMIC-IV baselines find fine-tuned language models merely competitive with tabular classifiers on one-shot prediction, with zero-shot models trailing both [22], and the single prospective deployment, an LLM adjudicating high-uncertainty sepsis alerts in silent mode, earns that role by augmenting a conventional predictor rather than replacing it [5].

Safety research, meanwhile, characterizes the failure modes that evaluation should be catching: taxonomies of medical hallucination grounded in clinician annotation [16], the case that clinical systems must communicate calibrated uncertainty rather than bare answers [17], threat models spanning an agent's reasoning, memory, and tools [50], and adversarial evidence that multi-agent topology determines how far a compromised agent's influence spreads [51]. Regulatory analysis converges from a third direction: general-purpose models already emit output meeting medical-device criteria [19], prompting proposed governance for unconfined non-deterministic clinical software built on guardrails, moderation, and inspectability [18]. Even the human backstop is less settled than assumed; the sole meta-analysis of clinician–LLM collaboration reports gains that are statistically fragile alongside non-trivial error rates in collaborative outputs [20].

Each literature names a requirement: grounded recommendations, calibrated confidence estimates, inspectable trails, structured oversight. To the best of our knowledge, none reports a system in which those requirements are implemented together and *measured* end-to-end on de-identified real-world patient records. That measurement, rather than any single component, is what the framework presented next is designed to make possible.

III. Methodology

A. Design Overview

The framework follows a design-science approach [52]: an artifact (the layered agentic framework) is constructed to close identified gaps and evaluated against explicit metrics. Six layers organize the system: data, memory, reasoning and knowledge, agent orchestration, clinical decision, and trustworthy AI. A clinician dashboard and human-in-the-loop validation close the loop, and a cross-cutting trustworthy-AI panel (explainability, audit logs, safety checks, bias monitoring, human oversight) governs every layer. Fig. 1 shows the full architecture over MIMIC-IV; this section states only the components that carry the paper's claims. Table I summarizes, per component, what is designed, what is implemented, and what has been evaluated, so that the boundary between the completed pilot (Section V) and the planned full evaluation (Section III-F) is explicit.

TABLE I
Component status: designed, implemented, and evaluated.

Component	Designed	Implemented	Evaluated
Patient-timeline memory (construction, management, retrieval)	Yes	Yes (non-LLM)	Pilot (Section V)
Timestamp-aware retrieval	Yes	Yes	Pilot: latency, no future da
Early-warning baseline	Yes	Yes	Pilot: AUROC with proxy
Verification gate (rule-based form)	Yes	Yes	Pilot: operating curve
Audit trail with measured resolvability	Yes	Yes	Pilot: 183/183 references
Seven LLM agents and coordinator	Yes	Scaffold only	Planned (Section III-F)
Guideline and literature RAG	Yes	No	Planned (Section III-F)
Calibrated confidence estimation	Yes	No	Planned (Section III-F)
Human-in-the-loop review protocol	Yes	UI prototype	Planned (Section III-F)
Longitudinal evaluation on full MIMIC-IV	Yes	No	Planned (Section III-F)

B. Patient-Timeline Retrieval

The memory layer treats the patient's longitudinal EHR trajectory as a first-class retrieval corpus alongside (not instead of) external medical knowledge. Four stores are maintained: short-term working context; long-term patient memory (admissions, diagnoses, medications, procedures); a vector database over clinical notes and events; and clinical context memory for the active monitoring episode. Memory operations follow the construction, management, and retrieval decomposition of the agent-memory literature [9]. Writes are explicit events linked to source records. Management handles decay and revision, so a corrected lab value supersedes but does not erase its predecessor. Retrieval is timestamp-aware, which keeps the evidence set limited to what was knowable at decision time. Retrieval-augmented generation then draws on both corpora, the patient timeline and guideline or literature knowledge [7], [43], and every recommendation can therefore cite patient-specific evidence rather than population-level knowledge alone [14], [15].

C. Agent Orchestration

This subsection describes designed, not demonstrated, behavior: as Table I records, the seven agents exist only as an implementation scaffold (typed interfaces, message schemas, and placeholder reasoning stubs), and no LLM-based reasoning ran in the pilot of Section V, whose intelligence resided entirely in a logistic-regression baseline and a rule-based gate. In the full system, a coordinator agent will delegate to seven specialized agents (monitoring, planner, diagnosis, risk prediction, treatment recommendation, explanation, and verification), each of which will use ReAct-style reasoning internally [6]. Coordination will follow a hub-and-spoke topology: all inter-agent messages will pass through the coordinator, a choice motivated by adversarial evidence that multi-agent topology governs how far a compromised agent's influence spreads, with open shared-communication designs proving most vulnerable to contamination [51]; routing every message through one auditable point will also serve the audit-trail requirement of Section III-D. Tool access will use MCP-style standardized interfaces and inter-agent exchange will follow A2A-style conventions [30], keeping the prototype composable with hospital tooling. External tools will follow the validated-tool principle of RiskAgent and TxAgent: calculations and lookups will be delegated to auditable components rather than generated [2], [1].

D. Verification Gate and Audit Trail

The verification agent is the framework's distinguishing safety component; only its rule-based form has been implemented and exercised (Section V), and the LLM-based form described here is designed, not built. In the full system, every candidate recommendation will be checked on three axes: (a) evidential grounding, i.e. whether the claim is entailed by the retrieved patient evidence and guidelines, screened against known classes of medical hallucination [16]; (b) guideline compliance and conflict with active orders; and (c) a calibrated confidence estimate that will accompany the recommendation to the clinician [17], [53]. Recommendations failing any check will be blocked and returned with the failure reason. Every check will write to an evidence-linked audit trail. Whether that trail accurately reflects the evidence the system actually used will be treated as a measured property rather than an assumed one, using entailment-based spot audits [54]. The confidence value attached to each recommendation will be calibrated post hoc on a held-out calibration split (temperature scaling, with isotonic regression as a fallback for non-monotone miscalibration) and its quality will be evaluated by expected calibration error and reliability diagrams [53]; no output of the current pilot is calibrated. This combination is designed to align with emerging regulatory expectations for unconfined non-deterministic clinical software: guardrails, moderation, retrieval grounding, and inspectability [18], [19].

E. Human-in-the-Loop Protocol

In the full system, clinicians will receive only verified recommendations, each with linked evidence, a calibrated confidence estimate, and an explanation. Review will be structured rather than free-form: approve, modify, or reject, with reasons captured into memory. The choice reflects meta-analytic evidence that unstructured clinician–LLM collaboration produces gains too uncertain to build on [20]. Rejections and modifications will feed the monitoring loop as supervision signals.

F. Planned Evaluation Design

This subsection specifies the full evaluation of the LLM-based framework; it is designed but has not yet been executed. The only completed empirical work in this paper is the non-LLM pilot reported in Section V, and no result below should be read as obtained. The planned full evaluation uses MIMIC-IV [13] cohorts (sepsis and deterioration use cases; Sepsis-3 labels [55], [56]) and scores the system on longitudinal tracking, not one-shot prediction. Clinical ground truth is defined concretely per use case: in-hospital mortality from discharge disposition, sepsis onset time under the Sepsis-3 criteria applied to the recorded observations [55], and, for treatment recommendations, concordance with the interventions documented in the record, adjudicated by clinician review where the record is ambiguous. The primary metrics are defined as follows. (1) Decision quality: discrimination (AUROC) and calibration of predicted risk against these labels, compared with revisited MIMIC-IV baselines [22]. (2) Grounding rate: of the atomic claims extracted from a recommendation, the fraction entailed by at least one retrieved patient-evidence item, following RAG-evaluation practice [54]. (3) Verification-gate effectiveness: among recommendations independently judged ungrounded, the fraction the gate blocks (catch rate) at a fixed clinician-review budget. (4) Audit-trail faithfulness: the fraction of logged evidence references that both re-resolve to the source record and match the value the system actually used, under human spot audit of a random sample sized to bound the faithfulness estimate within ś5 percentage points at 95% confidence (approximately 385 references, or the full trail when smaller). (5) Recommendation quality: rubric-graded scores in the style of physician-rubric benchmarks [21], [12]. Ablations remove patient-timeline retrieval, the verification gate, and longitudinal memory in turn, isolating each claimed contribution. Statistical comparisons follow standard practice for correlated classifiers [57], [58] with false-discovery control [59].

IV. Discussion

A. What the Framework Claims and What It Does Not

The framework's novelty is deliberately narrow. Tool-using medical agents exist [1], [2], [38]; multi-agent coordination exists [8], [28]; clinical RAG exists and works within its grounding assumptions [14], [15]; and longitudinal grounding itself is emerging in trajectory and guideline-alignment systems [44], [45], [47]. We do not claim any of these components. What we claim is their integration, and the instrumentation of that integration: timestamp-aware retrieval from the patient's own timeline coupled to recommendation-level verification, an audit trail whose faithfulness is measured rather than assumed, and evaluation that scores longitudinal tracking on de-identified real-world ICU records. Within the literature reviewed for this study, no system combines these, and none measures them together. If the ablations show these components do not improve grounded decision quality, the framework's thesis fails honestly; the components are separable and testable by design.

B. Relation to Contemporary Evaluation Practice

Contemporary benchmarks report two findings we take as fixed points: agents placed in a virtual EHR with physician-authored tasks succeed on only about seventy percent of them [12], and rubric-based grading of open-ended health conversations exposes quality variation that multiple-choice scoring conceals [21]. Our reading of those results, which the benchmark authors do not themselves draw, is that the remaining distance to bedside readiness is a substrate problem as much as a capability problem: synthetic records and scripted tasks cannot exhibit the revision, contradiction, and accumulation that make real admissions hard. Our evaluation design therefore keeps their instruments, task framing and rubric grading, while changing the substrate to real longitudinal MIMIC-IV admissions [13], [22]. The trade-off runs the other way too. MIMIC-IV is retrospective and single-region, and no retrospective design can establish the prospective safety that has so far been demonstrated for exactly one narrow task

[5]. We position retrospective longitudinal evaluation on real records as the missing rung between exam benchmarks and prospective trials, not as a substitute for the trials themselves.

C. Safety, Oversight, and Regulation

Each safety mechanism in the framework answers a failure mode the literature has documented rather than one we hypothesize. Clinician-annotated evidence that hallucinated medical content reaches practice [16] is why the gate screens recommendations before delivery; the argument that clinical systems owe users calibrated uncertainty, not bare answers [17], is why the design attaches a calibrated confidence estimate to every recommendation (planned, not implemented in the pilot; Table I); adversarial findings that open shared communication channels are the most vulnerable multi-agent design [51] are why the framework eliminates side channels and routes every message through one auditable point; and the demonstration that unconstrained model output already meets medical-device criteria [19] is why we treat regulatory alignment as a design input. We read the meta-analytic evidence on clinician–LLM teaming [20] as supporting a stronger conclusion than its authors state: oversight only earns its safety role when it is structured, which is why review in this framework is a protocol with mandatory reasons rather than an optional glance. Mapping the guardrails onto proposed UNDCS requirements [18] is our attempt to make a research prototype legible to that regulatory conversation; we expect the mapping, not the architecture, to need revision as the rules mature.

D. Limitations

Four limitations bound the claims, beyond the safety consideration that Section IV.5 treats separately. First, empirical evidence so far is a pilot on the 100-patient MIMIC-IV demo without the LLM loop (Section V); the full retrospective evaluation is designed but not executed, and no claim of prospective clinical benefit is made. Second, MIMIC-IV's critical-care population limits generalization to other care settings [13]. Third, verification quality is bounded by the entailment methods used to measure it [54]; a verification gate can only be as trustworthy as its own evaluation, which is why trail faithfulness is spot-audited by humans as well. Fourth, LLM API costs and latency constrain the monitoring cadence the prototype can sustain, a practical ceiling the agent-evaluation literature identifies as generally under-reported [49].

E. Safety Consideration: The Gate's Sensitivity Floor

The pilot's most consequential finding for deployment is not the gate's specificity but its sensitivity floor, and it warrants treatment as a first-class safety issue rather than a footnote to the results. At the strictest informative threshold the rule-based gate blocked three of the seven true alerts; a deployed system that suppressed half of the alerts preceding death would be clinically unacceptable, whatever its false-alert suppression. Three design consequences follow. The evidence window and signal threshold are safety parameters whose values clinical governance, not engineering, must set. Gate operating points must always be reported with sensitivity alongside false-alert suppression, as the pilot's operating curve does, so the trade-off is chosen deliberately rather than inherited from a default. And suppression must never be silent: in this framework blocked alerts remain visible and overridable in a dedicated queue rather than being discarded, so the gate defers, rather than replaces, clinical judgment on the alerts it declines to forward. The full evaluation will report sensitivity at every operating point, and we regard any deployment decision that has not confronted this floor explicitly as unsafe.

F. Future Work

Beyond executing the evaluation of Section III-F, three directions follow. Prospective shadow-mode deployment in the style of the deployed sepsis system is the natural next evidence level [5]. On-premises open models [35] would address the privacy constraints that hosted LLMs impose on real EHR timelines. And the verification gate invites formalization: moving from entailment heuristics toward auditable, possibly symbolic, checking of clinical claims against structured evidence, a direction the trustworthy-agent literature identifies as open [50], [18].

V. Pilot Feasibility Study

A. Rationale and Scope

The full evaluation of Section III-F requires credentialed MIMIC-IV access and an LLM agent loop; both are in progress. To establish that the framework's distinguishing components are implementable and behave as designed on de-identified real-world ICU data, we conducted a bounded pilot on the openly licensed MIMIC-IV Clinical Database Demo (v2.2; 100 patients, 275 admissions, 140 ICU stays) [13]. The pilot exercises the non-LLM slice of the architecture (timeline construction, timestamp-aware retrieval, an alert-generating classical baseline, and the verification gate with its audit trail) and is reported as feasibility evidence. With twenty deaths in the cohort, no performance claim survives this sample size, and we make none.

B. Setup

For each ICU stay, the memory layer assembled a chronological timeline from vitals, laboratory results, medication starts, and strictly prior admissions, each event carrying a provenance reference to its source table and row. Retrieval at a decision point (24 h after ICU admission) scored events by recency, type, and deterioration relevance, and never returned an event recorded after the query time. A standardized logistic regression over first-24 h features (5-fold stratified cross-validation) produced an in-hospital mortality risk score, and the top quintile generated 28 alerts. We define a *true alert* as an alert raised for a stay whose hospital admission ended in in-hospital death (7 alerts) and a *false alert* as one raised for a patient discharged alive (21 alerts). In-hospital mortality is a proxy endpoint chosen because it is unambiguous in the demo tables, not a claim that every deteriorating patient dies; much of monitoring's clinical value lies in non-fatal events such as escalation of care, fluid management, or antibiotic titration, which a mortality proxy cannot see. The planned full evaluation therefore replaces this proxy with Sepsis-3 onset and intervention-concordance labels (Section III-F). Leakage was controlled at three points: features were computed only from events time-stamped within the first 24 h of the ICU stay, while the mortality label derives from the discharge disposition recorded at the end of the admission; prior admissions entered the timeline only if discharged strictly before the current admission; and imputation and standardization were fit inside each cross-validation training fold via a pipeline, never on the full cohort. The verification gate passed an alert only when the patient's retrieved timeline contained at least m distinct deterioration-relevant signals (tachycardia, hypotension, tachypnea, hypoxemia, temperature derangement, elevated lactate, leukocytosis or leukopenia, rising creatinine or bilirubin, thrombocytopenia) within the six hours preceding the decision. Every gate decision logged its evidence references, and trail faithfulness was measured by re-resolving each logged reference against the raw tables.

C. Results

Timeline construction processed all 140 stays in 2.9 s (median 438 events per stay, IQR 236–798); 43% of stays carried at least one prior admission, confirming that longitudinal context exists to retrieve even in a small cohort. Retrieval answered queries in 11.7 ms median (17.9 ms p95) with full top-k coverage and returned no future information by construction. The baseline reached a cross-validated AUROC of 0.641 (95% bootstrap CI 0.478–0.792). Because the interval crosses 0.5, the baseline demonstrates no statistically significant discrimination at this sample size, and we draw no decision-support performance conclusion from it whatsoever. Its only function in the pilot is to generate a realistic, imperfect alert stream against which the gate's behavior can be observed.

The gate produced the pilot's most instructive result, with an essential scoping caveat: everything the gate demonstrates here is demonstrated on a stream of alerts, not on a stream of validated predictions, since the baseline generating those alerts has no established predictive validity. As the required number of distinct recent signals rises from 1 to 6, the pass rate for false alerts falls from 0.90 to 0.05 while the pass rate for true alerts falls more slowly (0.57 to 0.14); at $m = 4$ the gate blocks 86% of false alerts while retaining 43% of true ones. Evidence-gating therefore preferentially suppresses unsupported alerts within this stream, which is the mechanism the framework relies on. The same experiment exposes the mechanism's cost: three of the seven true alerts had no abnormal signal in the six-hour window at all, so the gate imposes a sensitivity floor governed by the evidence window, a safety parameter to be tuned, and an effect invisible in exam-style evaluation. All 183 logged evidence references re-resolved exactly against the raw tables (trail resolvability 1.00). In this deterministic pilot the perfect score is expected and carries no reliability claim: what the exercise benchmarks is the metric and its measurement machinery, not the system, and the metric earns its place because it becomes non-trivial once a non-deterministic LLM writes the trail.

D. What the Pilot Does and Does Not Show

The pilot shows that the framework's memory, retrieval, and verification mechanisms are implementable on de-identified real-world ICU records at interactive latency, that evidence-gating discriminates in the intended direction, and that audit-trail faithfulness is a measurable quantity. It does not show clinical utility, generalization beyond a 100-patient demo cohort, or anything about the LLM agent loop, which the full MIMIC-IV evaluation of Section III-F will test. The demo lacks the clinical notes module, so retrieval here is structured-data only.

The relation between the pilot gate and the framework's verification agent deserves explicit statement. The pilot gate is a deliberately simple heuristic instantiation of the verification interface: it consumes the same retrieved-evidence set, emits the same pass-or-block decision, and logs the same evidence references that the LLM-based agent of Section III-D will. What changes in the full system is the decision rule, from counting distinct deterioration signals to claim-level entailment against the retrieved evidence. Whether the preferential suppression of false alerts observed here transfers to the entailment-based form is therefore an explicit hypothesis for the full evaluation, not an assumption this pilot licenses.

VI. Conclusion

Medical LLM agents can now act inside clinical workflows; whether their actions can be trusted for a specific patient over time is a question the systems reviewed in Section II neither answer nor measure. This paper's response is architectural and methodological rather than model-centric. The framework's distinguishing commitments (the patient timeline as a retrieval corpus, a verification gate that checks recommendations against retrieved patient evidence, an audit trail whose faithfulness is measured, and structured rather than advisory clinician oversight) each correspond to a requirement that the benchmark, safety, and regulatory literature cited throughout names but does not yet report measured together in one system. The pilot on the open MIMIC-IV demo establishes a bounded but concrete result: the non-LLM substrate of these commitments runs on de-identified real-world ICU records at interactive latency, and requiring recent, concordant, multi-signal evidence blocks unsupported alerts at a substantially higher rate than supported ones, while also exposing a sensitivity floor that examination-style evaluation cannot reveal. The pilot's limits are just as clear. It says nothing about the LLM agent loop, whose test is the full MIMIC-IV evaluation specified in Section III-F, and nothing about clinical utility, which only prospective study can establish. If the full evaluation shows these components fail to improve grounded decision quality, the design permits that failure to be located exactly. We therefore consider measurable verification and traceability essential properties for evaluating clinical agentic AI.

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